

Relatedness in PCA

and how to deal with it

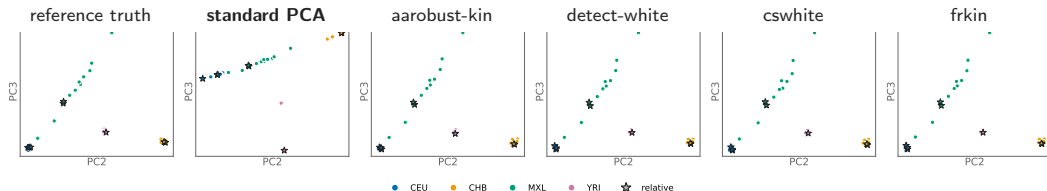
PCAone --robust: small samples, biobanks and admixed relatives

relatePCA

October 5, 2026

Relatives add covariance that is not ancestry

- PCA of genotypes is a PCA of their covariance between individuals.
- Relatives share alleles identical by descent: $\text{Cov}(g_i, g_j) = 4\phi_{ij} \pi(1 - \pi)$, on top of ancestry.
- A family is a small, tight cluster: PCA spends components on it.
- Goal: **every individual in-sample** (no projection), PCs not shaped by close relatives (2nd degree and closer).



PC2 vs PC3; admixTjck2, 10 per population + 8 relatives (stars). Standard PCA loses the 3rd axis (min $R^2 = 0.013$ against the reference truth); every robust mode recovers it (≥ 0.997).

Two regimes

Small samples (a few individuals per population)

- one family can capture an ancestry axis;
- standard PCA fails in 79% of benchmark runs ($\min R^2 < 0.95$).

Large samples

- the top axes survive;
- relatives are misplaced;
- the PCs just beyond the ancestry axes become *family axes*.

$N = 2000$, 30% in families	standard	robust
top 4 axes, $\min R^2$	0.956	0.957
relatives' error	0.49	0.12
family axes (of 16 extra PCs)	16	0
largest family share η^2	0.91	0.17

Simulated, 5 populations (F_{ST} 0.05–0.002) + admixed; truth: PCA of a separate reference panel.

Small N: an accuracy problem. Large N: a placement and extra-PC problem.

The usual fixes and what they cost

- **Remove relatives** (keep one per family): loses individuals and their PCs; needs a pedigree or a relatedness screen first.
- **PC-AiR**: PCA of an unrelated subset, relatives *projected*: relatives are not in-sample, and projection shrinks them.
- **KING-robust screen**: fine within an unadmixed population, biased across populations and for admixed individuals (later).

What we want instead

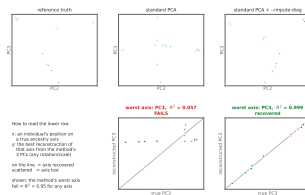
- a decomposition of the individual-by-individual matrix into *structure* (low rank) and *relatives* (sparse);
- a kinship threshold, not a tuning constant: $\tau = 2^{-3.5} \approx 0.088$, between 2nd (0.125) and 3rd degree (0.0625);
- PCs from the structure, or from a matrix in which families are *whitened* (down-weighted to their effective size).

Small N I: the diagonal

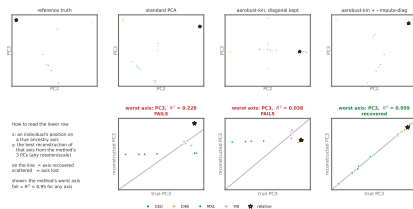
- The GRM diagonal is a structure part plus heterozygosity noise.
- At $n = 5$ per population the noise dominates, and it differs by population (YRI highest).
- The uneven diagonal acts as a population indicator: PCA spends an axis on it — *even without relatives*.
- Fix: treat the diagonal as **unobserved**, impute it from the off-diagonal entries (`--impute-diag`).

At $n = 5$, mean min R^2 : diagonal kept 0.022 (unrelated only) / 0.018 (with relatives); imputed 0.985 / 0.989.

A. 20 unrelated individuals (5 per population)



B. the same 20 plus two MZ twins (stars)



Small N II: centring leaks the diagonal into every entry

- The GRM is built from centred genotypes:
 $C = JAJ$, $J = I - \mathbf{1}\mathbf{1}'/N$.

- A diagonal noise term becomes

$$JDJ = D - \frac{d\mathbf{1}' + \mathbf{1}d'}{N} + \frac{\bar{d}}{N}\mathbf{1}\mathbf{1}'$$

so C_{ij} picks up $-(d_i + d_j)/N$.

- Rows of C sum to zero: freeing the diagonal frees nothing.
- Undo it: $A = C + u\mathbf{1}' + \mathbf{1}u' + (\mu'\mu/M)\mathbf{1}\mathbf{1}'$,
 $u = X_c\mu/M$ — needs f and one number per individual.

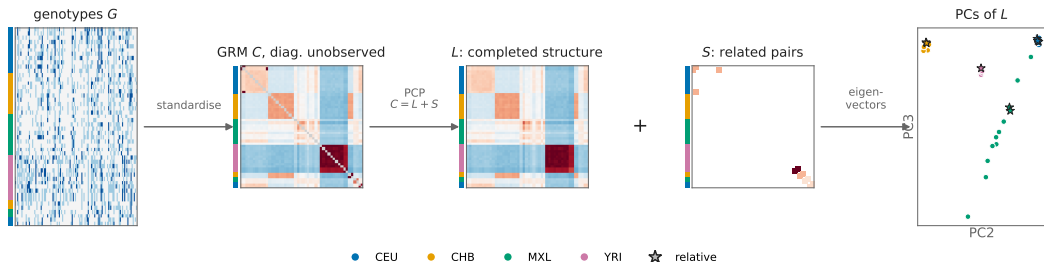
rank fit, diagonal free, on	failures, $n = 5$
GRM, scaled + centred	82%
GRM, scaled, not centred	0%
CS matrix (raw, not centred)	0%
PCP on the centred GRM	2.5%

Scaling by $2f(1 - f)$ makes no difference; centring does.

PCP survives only because its fit ends up full rank.

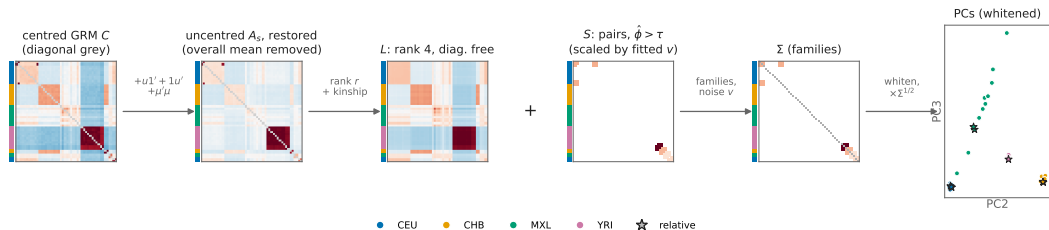
Work on the uncentred matrix (or restore it), with the diagonal free.

Robust PCA of the GRM with a kinship threshold (aarobust-kin)



- PCP $C = L + S$: diagonal always in S ; a pair enters S only if its kinship exceeds τ (and it passes a screen) — no λ , no rank.
- Needs the full spectrum in every iteration (N^3): 31 s at $N = 1000$, 248 s at $N = 2000$. Partial (IRAM) solves do not help: L ends up full rank.

dwg: detect, then whiten (the default)



- Uncentred, SNP-standardised Gram; low-rank fit with the diagonal free; kinship scaled by the *fitted* noise v_i (no HWE within individuals).
- Whitening $\Sigma = v^{1/2}(I + 2\Psi)v^{1/2}$: a family counts as its effective number of individuals. Only the top r eigenvectors per step.

Choosing the rank without knowing K

- Raise the rank while the next eigenvalue exceeds the noise edge $2\sigma\sqrt{N}$; k is only an upper bound.
- **Family axes:** an eigenvector resting on < 4 individuals ($n_{\text{eff}} = 1/\sum u_i^4$) with a related pair among them is a family, not structure: its pairs become candidates, the rank is not raised.
- Ancestry axes rest on whole populations (n_{eff} 14–49 at $n = 20$), family axes on 2–3 individuals.

	KING screen	dwg
failures $k = 10, n = 5$	6.7%	1.7%
failures $k = 10, n = 10$	6.7%	0%
relatives' error ($k = 10$)	0.25	0.09
runs changed by k (of 240)	12	2

12 scenarios incl. relatives of different ancestry, 5 replicates;
 $k = 3$ and $k = 10$ with true $K - 1 = 3$.

Whiten, do not just impute

- Alternative: PCs of the Gram with the related pairs and the diagonal imputed. As accurate on the top PCs, and all k PCs exist.
- But imputation leaves each relative a full individual: sibs keep near-identical correlations with everyone else.
- Whitening corrects the weighting.

$N = 2000$, PCs 5–20	family axes	median η^2
standard PCA	16 of 16	0.92
imputed pairs + diagonal	0	0.37
whitening	0	0.21

η^2 : share of a PC's variance explained by family membership (30% of the sample in families).

Admixed I: KING is not structure-free

- KING-robust corrects for heterozygosity differences between populations: valid for two individuals from the *same unadmixed* population.
- Relatives of different ancestry: pulled **down**.
- Admixed individuals of similar ancestry: pulled **up**.
- Below the 0.04 screen a pair can never be called related.

true pair (kinship)	KING	evalAdmix
grandparent – grandchild ^a (0.125)	−0.013	0.085
aunt – nephew ^b (0.125)	0.017	0.089
half-sibs ^c (0.125)	0.069	0.089
parent – child ^d (0.25)	0.187	0.236

^aYRI and $\frac{1}{4}$ YRI; ^bCHB and CHB/YRI; ^cCEU×CHB and CEU×YRI;
^dCEU×YRI child. Lowest values over 5 replicates; relatives from real admixTjeck2 genotypes with recombination.

KING-only screening missed 40–47% of these cross-ancestry 2nd-degree pairs.

Admixed II: individual allele frequencies

- Predict each genotype from the individual's own ancestry: individual allele frequency π_{is} ; relatedness is what the prediction leaves.
- PC-Relate, PCAngsd and evalAdmix are **one estimator**: the correlation of residuals estimates 2ϕ . They differ only in
 1. where $\hat{\pi}$ comes from (PCs or an admixture model),
 2. the denominator (model or observed variance),
 3. whether the pair's own genotypes were used to build $\hat{\pi}$.
- The number of PCs matters most: use $K - 1$ (plus intercept). Too many PCs absorb the relatives themselves.

RMSE	PC-Relate	evalAdmix (EM)	PCAone --evaladmix
unrelated pairs	0.00330	0.00378	0.00386
related pairs	0.00358	0.00292	0.00251

Admixed III: an admixture-aware screen in `dwg`

1. First fit with the KING screen (a safe guard for the early, low-rank steps).
2. `evalAdmix` kinship from the fit's *structure* axes (family axes left out) $\rightarrow$ candidates > 0.04 .
3. `evalAdmix` alone flags unrelated admixed pairs of similar ancestry at small N (two MXL: 0.113). Confirm by **IBD**: $k_0 < 0.8$ (unrelated ≈ 1 , 2nd degree ≈ 0.5).
4. Refit; stop when the pairs no longer change.

recall, $k = 10$	KING	<code>dwg</code>
$n = 5$	88%	95%
$n = 10$	88%	96%
$n = 20$	86%	97%
$n = 40$	90%	100%
false pairs	0	0

Same accuracy as an `evalAdmix` screen with the true $K - 1$
— without knowing K .

Admixed IV: final relatedness, (k_0, k_1, k_2)

- Kinship alone cannot separate parent–offspring from full sibs (both $\frac{1}{4}$); k_0 can (0 vs $\frac{1}{4}$).
- Allele-level likelihood with $\pi_i \neq \pi_j$: pooling the two frequencies reports a parent–offspring pair as 83% unrelated.
- Allele frequencies from the final PCs, the pair's **whole family left out**; projected Newton (EM stalls at $k_0 = 0$).

$N = 2000$ (simulated)	kinship	k_0	k_1
parent–offspring	0.251	0.000	0.996
full sibs	0.250	0.250	0.500
2nd degree	0.126	0.504	0.490
MZ	0.500	0.000	0.000

Written to .relpairs. At $N = 47$ (real genotypes): PO and sibs separated; 2nd degree ≈ 0.02 too high.

Real LD and real panels

HAPNEST (synthetic biobank, realistic LD, 6 ancestries), $N = 5000$

	standard	detect-white	dwg
min R^2 (5 axes)	0.9987	0.9977	0.9988
relatives' error	0.048	0.028	0.019
family axes	15/15	0	0
recall	–	97.3%	98.3%
cross-anc. grandparent	–	66%	87%
time	14 s	100 s	21 s

1000 Genomes (CEU, CHB, YRI, ASW), $n = 5-20$

	failures	recall
standard	27% / 10% / 0	–
aaobust-kin	0	78–84%
detect-white	0	78–83%
dwg	0	90–96%

Every pair **dwg** missed on HAPNEST had realised kinship $< \tau$. The extra 1000G pairs are real relatives in the panel (e.g. NA20317/NA20318).

Large N : no $N \times N$ matrix

- Operator engine: each iteration is one pass $G(G'V)$ with a few vectors; in-core or out-of-core.
- Candidates: bit-packed KING over all pairs (popcounts); exact GRM-scale entries for them in one weighted pass.
- Family-axis checks by one product with unit vectors.

16 threads	dwg	standard
$N = 1000$	1.1 s	–
$N = 5000$	4.7 s	3.5 s
$N = 20,000$ in-core	26.5 s	24 s
$N = 20,000$ out-of-core	109 s	29 s

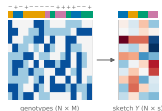
Dense engine (dwg): 10.7 s at $N = 2000$; aarobust-kin 248 s.

Very large N : finding the relatives with a sketch

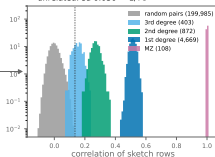
Very large N ($N > 20,000$): sketch search, then dwg

--robust (auto); --king-search auto

1 count sketch in pass 1: SNP j to column $h(j)$ with sign $\pm$



2 relatives stand out ($N = 20,000$, $s = 2048$)
unrelated: $\text{sd } 0.030 \approx 1/\sqrt{s}$



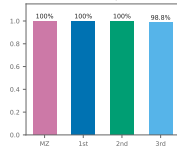
3 nearest neighbours

top $m = 5$ per person:
blocked float products,
cost N^2s , not N^2M

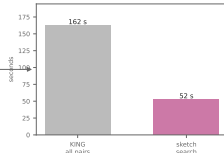
all m are relatives?
search $2m$, $4m$, ...

exact KING-robust on
~3.5 candidates / person

4 recall vs all-pairs KING
($N = 20,000$; same pairs at 100k)



5 finding the relatives
 $N = 100,000$, 16 threads



6 then as for large N

dwg fit, residual-sketch
evalAdmix + k_0 screen,
whitening (operator engine)

final relatedness:
 ϕ , k_0 , k_1 , k_2 (1 pass)

$N = 100k$: dwg 202 s
(detect-white 118 s;
standard PCAone 148 s)

Sketch: $N \times M$ (columns and signs from a hash of the SNP index; in-core = out-of-core). Search: N^2s , independent of M .
Relatives: correlation about 2ϕ ; unrelated: $0 \pm 1/\sqrt{s}$ (dotted: expected maximum over N). 1-bit signs and LSH bucketing do not work.

Count sketch in the first pass ($N \times M$); nearest neighbours N^2s , independent of M ; exact KING on ~3.5 candidates per person. $N = 100,000$: the same 27,880 pairs as all-pairs KING, 52 s instead of 162 s.

Genotype likelihoods (-G)

- Posterior means shrink towards the prior — more at low depth.
- **Deshrink**: $x = 2f + (E - 2f)/\rho_i$, ρ_i the genotype reliability; off-diagonal products $\approx$ unbiased, extra noise on the free diagonal.
- Kinship decisions $/\sqrt{\rho_i\rho_j}$; screen with KING from expected counts; k_0 and final (k_0, k_1, k_2) from the likelihoods.
- One round with the HWE prior beats iterated (PCAngsd-style) priors.

failures	$n = 10$	$n = 5$
true genotypes	0	0
GL , 8 $\times$	0	6%
GL , 4 $\times$	0	12.5%
GL , 2 $\times$	0	19%
standard GL PCA, 4 $\times$	100%	100%

Simulated reads (1% error) on admixTjeck2, 54k SNPs. At 1 $\times$ nothing works with 5 per population.

Using it

```
PCAone -b data -k K-1 --robust          # dwg, any N
PCAone -b data -k 10 --robust -m 2       # out-of-core
PCAone -G data.beagle.gz -k K-1 --robust # genotype likelihoods
PCAone -b data -k 3 --robust --impute-diag # aarobust-kin (PCP)
PCAone -b data -k K-1 --robust --kinship pairs.kin0 # known pedigree
```

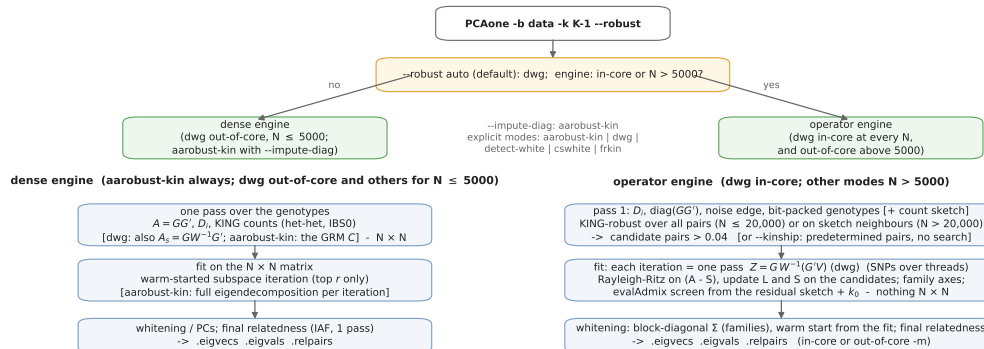
- `.eigvecs` / `.eigvals`: the robust PCs (a generous k is safe).
- `.relpairs`: KING, KIN_DETECT (decides the pair), KINSHIP K0 K1 K2 (final, from the final PCs), KIN_EVALADMIX.
- Code and the full methods document: `branch robust-relatives, docs/robust/README.md`.

Summary

- Relatives distort PCA differently by sample size: a lost axis at small N , misplaced relatives and family axes at large N .
- At small N the GRM diagonal is a second, independent problem — and centring spreads it into every entry. Work uncentred, diagonal free.
- `dwg`: detect related pairs with a kinship threshold, then whiten. No HWE assumption, k only an upper bound, all individuals in-sample; from 20 individuals to 100,000.
- Admixed relatives: KING misses them; an `evalAdmix` screen from the structure axes, confirmed by IBD, finds them.
- Final relatedness from the final PCs: (k_0, k_1, k_2) with $\pi_i \neq \pi_j$ and leave-the-family-out frequencies.

Open: missing genotypes; parent–offspring k_2 with low-depth GL; first cousins at ~ 5 per population.

Backup: the three regimes and the engines



Backup: building blocks

- KING-robust: $\phi_{ij}^{\text{KING}} = \frac{N_{ij}^{\text{het,het}} - 2N_{ij}^{\text{IBS0}}}{2m_{ij}} + \frac{1}{2} - \frac{N_i^{\text{het}} + N_j^{\text{het}}}{4m_{ij}}$, $m_{ij} = \min(N_i^{\text{het}}, N_j^{\text{het}})$.
- Structure-adjusted kinship from a residual $R = A - L$: $\hat{\phi}_{ij} = R_{ij} / (2\sqrt{v_i v_j})$.
- Noise edge: $2\sigma\sqrt{N}$, $\sigma^2 = \sum_s (E[g^2]^2 - (2f_s)^4) / (w_s^2 M^2)$.
- IBD likelihood per site (allele level): IBD0 $g_i \sim \text{HWE}(\pi_i)$, $g_j \sim \text{HWE}(\pi_j)$; IBD1 shared allele $\sim \frac{1}{2}(\pi_i + \pi_j)$; IBD2 $g_i = g_j \sim \text{HWE}(\frac{1}{2}(\pi_i + \pi_j))$.